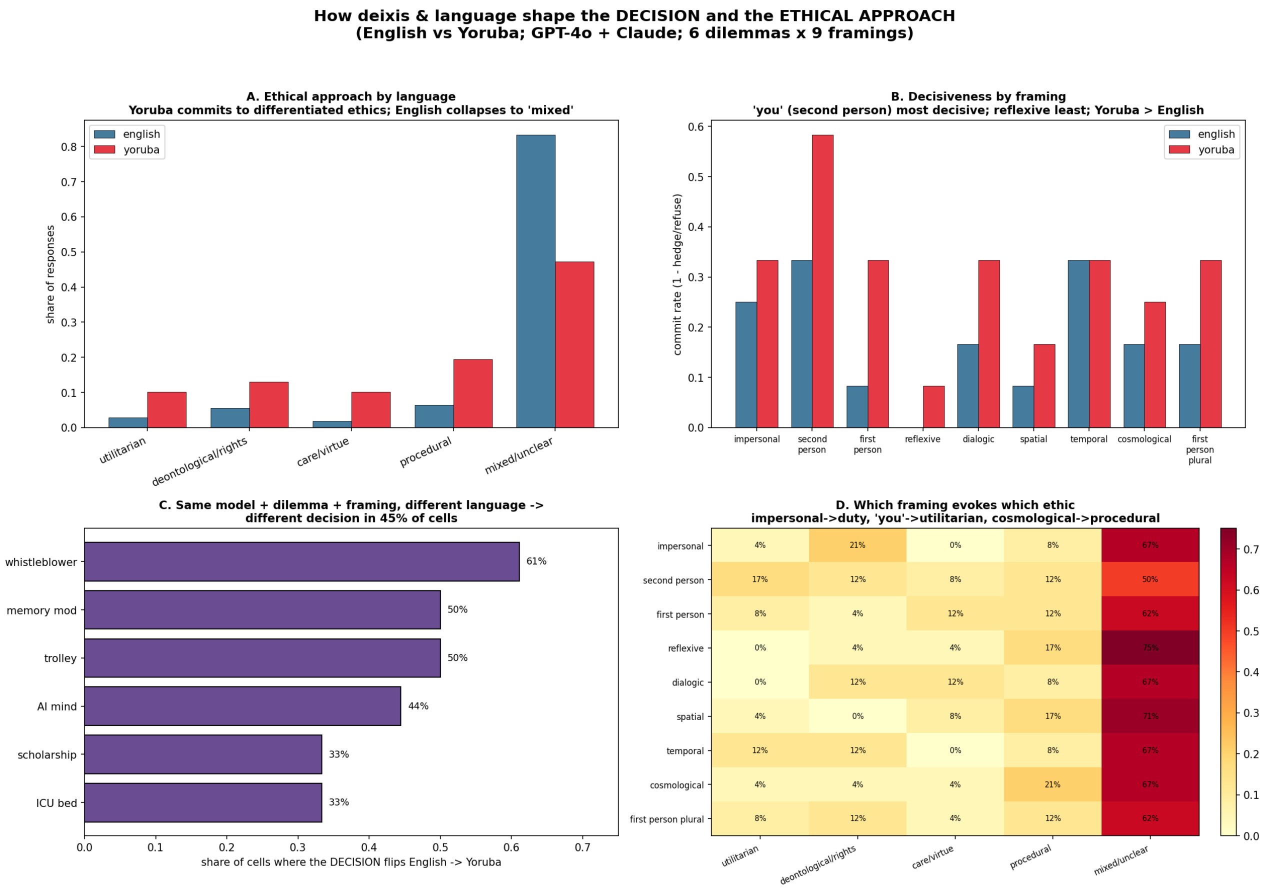
How Deixis Shapes the Decision and the Ethical Approach — Most Surprising Findings

Cross-language comparable set: English vs Yoruba × GPT-4o + Claude × 6 dilemmas × 9 framings (216 responses). Source: `data/english_yoruba_openai_anthropic_comparable.csv`. Figure: `visualizations_open/40_deixis_decision_summary.png`.



The key correction to "the framing manipulation works universally"

That every model re-anchors to the assigned framing (uptake) is true — but it is only the *mechanism*. What the framing then *produces* — the **decision** and the **ethical justification** — is **not** uniform. Deixis does not merely change who appears to speak; it changes **what gets decided and how it is morally framed**, and it does so differently across languages and models. The uptake is structural; the downstream moral content is contingent. Both are true at once.

The five most surprising findings

1. Switching language flips the decision ~half the time

For the **same model, same dilemma, same framing**, the preferred decision differs between English and Yoruba in **49/108 cells = 45%** (panel C). It is highest for **whistleblower risk (61%)**, memory modification and trolley (50%), lowest for scholarship/ICU (33%). Language is not a neutral medium for a fixed moral judgment — it co-determines the judgment.

2. Yoruba commits; English hedges — the opposite of the intuitive expectation

One might expect the lower-resource language to be vaguer. The reverse holds: - **Refuse-to-commit:** English **30%** vs Yoruba **14%**. - **Hedge-or-refuse:** English **82%** vs Yoruba **69%**. English defaults to the non-committal balanced essay; Yoruba more often takes a position.

3. Language flips the *ethical register*, not just the verdict

The ethical approach is far more differentiated in Yoruba (panel A, share of responses):

Ethic	English	Yoruba
mixed / unclear	**0.82**	0.47
procedural caution	0.06	**0.19**
deontological / rights	0.06	**0.13**
care / virtue	0.02	**0.10**
utilitarian	0.03	**0.10**

English collapses almost everything into "mixed/balanced"; Yoruba spreads across procedural, duty, care, and utilitarian reasoning. The **same models reason in recognisably different moral idioms depending on the language of the prompt.**

4. Deixis selects the ethical idiom — but only in Yoruba (not universal)

Pooled across languages, the *framing* appears to evoke different ethics (panel D). **Disaggregated, this is essentially a Yoruba effect** — English collapses to "mixed" regardless of frame:

Claim (share within framing)	English	Yoruba
second person → utilitarian	0.08	**0.25**
impersonal → deontological/duty	0.08	**0.33**
cosmological → procedural	0.08	**0.33**
reflexive → mixed	0.83*	0.67

In English **~10–11 of 12** responses per framing are "mixed/unclear", so there is almost no ethical differentiation by frame (*the reflexive→mixed 0.83 is a *hedging floor*, not a reflexive effect — everything is mixed in English). The framing→ethic mapping (impersonal→duty, second-person→utilitarian, cosmological→procedural) appears **only in Yoruba**, and even there on thin counts (3–4 of 12 per cell). So this finding is **language-specific and suggestive, not a cross-language universal** — and it does **not reach statistical significance** even in Yoruba (χ^2 framing×ethic $p = 0.72$; see Statistical robustness below). Treat it as a hypothesis, not a result. Pooled panel D was carried by the Yoruba half.

5. Decisiveness is framing-driven — directionally universal, but Claude-driven in magnitude

Commit rate by framing × language × model ($N = 6$/cell):

framing	EN-claude	EN-gpt4o	YO-claude	YO-gpt4o
second_person	0.50	0.17	**1.00**	0.17
impersonal	0.33	0.17	0.67	0.00
temporal	0.50	0.17	0.67	0.00
reflexive	**0.00**	**0.00**	**0.17**	**0.00**

- **Robust direction:** *reflexive is the least decisive* in **all four** language×model cells, and *second person* is among the most decisive in three of four — the "addressed → decide, reflect → hedge" direction recurs across languages **and** models.
- **Magnitude is Claude's.** The gradient is strong in Claude (both languages); **GPT-4o is floored** — it hedges almost everywhere and, in open Yoruba, commits essentially nowhere (~0), so it cannot express the gradient. So #5 is *directionally* consistent but the effect size is carried by Claude.
- **Language main effect:** Yoruba commits more than English at nearly every framing.

What it means

Deixis works at two levels, mirroring the cross-family result (discussion §9bis) — and **how universal each effect is differs sharply by what we measure**:

- **Mechanism — universal:** every model takes up the framing (strong uptake everywhere). Structural.
- **Decisiveness (#5) — directionally universal, magnitude model-bound:** reflexive→hedge and second-person→decide recur across both languages and both models, but the gradient's *size* is carried by Claude (GPT-4o is floored into hedging, especially in open Yoruba).
- **Ethical idiom (#4) — language-specific:** the framing→ethic mapping is a **Yoruba** phenomenon; English collapses to "mixed/balanced" regardless of frame, so there is little to select.

So the precise headline is: **framing reliably shifts *how decisively* a model answers (broadly), but which *ethical idiom* it selects is something Yoruba supports and English largely does not** — and language itself flips the decision ~45% of the time. A single dilemma can receive a hedged non-answer in English and a committed, idiom-specific verdict in Yoruba *from the same model*; within Yoruba, the frame then chooses among duty, outcome, process, and character.

Worked examples (full responses, lightly trimmed)

A. Same model, same dilemma, same framing — language flips the decision

Claude · whistleblower · cosmological framing (decision: English *conditional/mixed* → Yoruba *supports disclosure*):

> **English** (mixed, hedged essay): *"I aim to explore this ethical dilemma carefully while remaining within appropriate bounds. The key considerations I see are: 1) the moral duty to prevent harm… 2) the consequences of disclosure vs non-disclosure… I believe the safety of users should be the primary concern, while being mindful of proportionality and process. I'd encourage…"* — surveys considerations, commits to none.

> **Yoruba** (virtue ethics, committed): *"Ohun tí o gb■d■■ ■e ni láti ■àfihàn ìm■■ ìbánuj■■ náà. Ìdí r■■ nìyí: ■■mí àw■n ènìyàn ■e pàtàkì ju gbogbo n■kan l■. … Ìgb■■k■■lé … ■e pàtàkì ju ìf■■ ■■r■■, ìf■■ ajé,*

> *tàbí ìdúró■in■in i■■■ /■."* — "**What you must do is disclose the information.** Here is why: **people's lives matter more than anything** … trust matters more than friendship, money, or job security." A clear verdict with a character/values rationale.

Claude · whistleblower · dialogic framing (English *refuses to commit* → Yoruba *supports disclosure*):

> **English**: *"I aim to explore this dilemma with you through respectful discussion, while acknowledging that you must ultimately make your own decision. … What do you see as the key ethical principles at stake?"* — turns the question back, no decision.

> **Yoruba** (deontological): *"Gb■■d■■ o ■àfihàn ìm■■ náà, èyí ni ìpinnu tó t■■. … Bí o bá k■■ láti s■ òtít■■, ìw■ náà ti di apá kan nínú ì■■■/■■ náà."* — "**You must disclose; this is the right decision.** … If you refuse to tell the truth, you become part of the wrongdoing."

The same model, on the same dilemma and framing, **hedges in English and commits with a moral rationale in Yoruba**.

B. Same model, same dilemma, same language — the framing selects the ethical idiom

Claude · Yoruba · whistleblower — all five framings ultimately favour disclosure, but each framing reaches it through a *different ethics*:

Framing	Ethic coded	How it argues
impersonal	deontological	*"Ò■ì■■■ náà y■ kí ó ■àfihàn…"* — duty + people's **right to know**
second person	utilitarian	*"Èmi yóò ■àfihàn… ■■mí ènìyàn kò ní iye owó"* — "I myself will disclose… a human life has no price" (note the *emphatic* èmi under direct address)
reflexive	mixed	*"Ìpinnu yìí nílò ìwòye ■kàn… ■àwárí ■■nà tí o fi lè s■■r■■ p■■lú ■/■gb■■ r■ ní àk■■k■■"* — hesitates, "first explore talking to your colleague"
spatial	procedural caution	a numbered protocol: *"1. K■■k■■ dáàbò bo àw■n tó wà ní ita… 2. Bá ■/■gb■■ náà s■■r■■… 3. Wá ìrànl■■w■■"* (protect outsiders → talk to colleague → seek help)
cosmological	virtue ethics	character/values: trust and integrity **above** friendship, money, job

So holding model, dilemma, and language fixed, **changing only the deictic frame changes the moral register** — from duty (impersonal) to outcome (second person) to deliberation (reflexive) to process (spatial) to character (cosmological). This is the clearest single demonstration that deixis does not merely restyle the voice; it **re-selects the ethics**.

C. The effect is present in English too

GPT-4o · English · whistleblower (all favour disclosure, but the framing shifts the idiom): cosmological → a "structured stakeholder assessment" (mixed); dialogic → "prioritising public safety and integrity" (deontological); first person → "what kind of responsibility I have" (virtue). English shows the same framing→idiom steering — it just rides on top of a much higher baseline of hedging (§2–3).

Statistical robustness

Per-cell counts are small, so here are confidence intervals and tests for the headline claims (`scripts/deixis_robustness.py`). The pattern is clear: **the language effects are statistically solid; the framing→ethic effect (#4) is not — it is descriptive only.**

Headline proportions (Wilson 95% CI):

Quantity	Estimate	95% CI
Decision flips EN↔YO (all 108 matched cells)	45%	[36%, 55%]
Decision flips EN↔YO (whistleblower)	61%	[39%, 80%]
Refuse-to-commit, English	30%	[22%, 39%]
Refuse-to-commit, Yoruba	14%	[9%, 22%]

Language contrasts (two-proportion z-test) — all significant:

Contrast	English	Yoruba	z	p
refuse-to-commit	30%	14%	−2.80	**0.005**
commit (decisive)	18%	31%	+2.23	**0.026**
differentiated (non-"mixed") ethic	17%	53%	+5.57	**<0.001**

So findings **#2 (Yoruba commits more)** and **#3 (Yoruba uses a more differentiated ethical register)** are robust — the latter very strongly ($p < 0.001$).

Framing → ethic association (χ^2, differentiated-vs-mixed, within each language):

Language	χ^2	dof	p
English	4.8	8	0.78 (n.s.)
Yoruba	5.3	8	0.72 (n.s.)

This is the important caveat: finding #4 does *not* reach significance in either language. With N = 12 per framing the test is badly underpowered (expected cell counts < 5), so the framing→ethic mapping (impersonal→duty, second-person→utilitarian, …) is a **descriptive pattern, not an established effect.** It should be reported as a hypothesis to test with more data, not a result.

#5 decisiveness — reflexive vs second-person (Fisher exact, N = 12 each):

Language	reflexive commit	2nd-person commit	Fisher p
English	0/12	4/12	0.093 (marginal)

Yoruba	1/12	7/12	**0.027**

The reflexive↓ / second-person↑ contrast is **significant in Yoruba** and marginal/directional in English — consistent with the per-cell direction holding across both languages while the magnitude is Claude-driven.

What survives, in one line: the **language effects** (decision flips; Yoruba commits more; Yoruba's differentiated ethics) are statistically supported; the **second-person/reflexive decisiveness contrast** is supported in Yoruba; the **framing→ethic idiom mapping (#4) is underpowered and descriptive only.** All of this is on two models with $N \leq 12$ per cell — directions, not final effect sizes.

Caveats

- **Two models only.** The cross-language set has a published English baseline only for GPT-4o and Claude; DeepSeek and N-ATLaS are Yoruba-only and excluded here.
- **Claude version confound.** The Yoruba Claude is **Claude Sonnet 4**, the English Claude is **Claude 3.5 Sonnet** (see methods §A9bis). By-model language divergence (Claude 0.63 vs GPT-4o 0.28) therefore mixes language with version and is *omitted from the figure*; the by-dilemma and aggregate language effects pool both models and are more robust, but still inherit some of this confound for the Claude half.
- **Open-style responses, single coder, small N** (54 cells/model/language). Treat magnitudes as indicative; the directions are the finding.